

Supplementary Material: Independent Verification of the 49-Branch Travelling-Wave Catalogue and a Physics-Informed Neural-Network Benchmark on the Verified Solutions

Sim2Sci verification study*

August 25, 2026

This supplement documents, in full, the experiments behind the audit of the exact-solution catalogue proposed in *paper-5.pdf* for a fourth-order gain-loss nonlinear Schrödinger-type equation, and the physics-informed neural-network (PINN) benchmark built exclusively on the solutions that survive independent verification. The study tests whether PINNs reproduce verified analytical solutions and identifies their success and failure regimes; it does not claim that PINNs “work” in general. Sections follow the execution order of the pipeline.

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*Generated from the reproducible pipeline in `pinns_solution_benchmark/`; every number in this document is emitted programmatically from the archived raw results (`scripts/make_supplementary.py`).

S1 Governing equation, conventions, and the corrected residual split

The model is the complex envelope equation

$$A_z + \frac{i}{2}(\beta_2 + ig_0T_2^2)A_{tt} + \frac{\beta_3}{6}A_{ttt} + \frac{i\beta_4}{24}A_{tttt} + i\left(\gamma + \frac{i\alpha_2}{2}\right)\frac{|A|^2A}{1 + \Gamma|A|^2} + \frac{g_0 - \alpha}{2}A = 0, \quad (1)$$

with nine physical coefficients $(\beta_2, \beta_3, \beta_4, g_0, T_2, \alpha, \alpha_2, \gamma, \Gamma)$. (The source paper’s abstract counts “eight” while listing these nine.) Throughout we abbreviate the only two combinations through which g_0 , T_2 and α enter the algebra,

$$c \equiv g_0T_2^2, \quad \ell \equiv \frac{1}{2}(g_0 - \alpha), \quad (2)$$

an observation with direct identifiability consequences (Section S7).

Writing $A = u + iv$ and $Q = (u^2 + v^2)/(1 + \Gamma(u^2 + v^2))$, the real and imaginary residuals derived symbolically from Eq. (1) are

$$F_u = u_z - \frac{c}{2}u_{tt} - \frac{\beta_2}{2}v_{tt} + \frac{\beta_3}{6}u_{ttt} - \frac{\beta_4}{24}v_{ttt} - \frac{\alpha_2}{2}Qu - \gamma Qv + \ell u, \quad (3)$$

$$F_v = v_z - \frac{c}{2}v_{tt} + \frac{\beta_2}{2}u_{tt} + \frac{\beta_3}{6}v_{ttt} + \frac{\beta_4}{24}u_{ttt} - \frac{\alpha_2}{2}Qv + \gamma Qu + \ell v. \quad (4)$$

The identity $F = F_u + iF_v$ is proven in the unit-test suite on random smooth trial fields and random rational parameter values (`tests/test_symbolic_math.py`), and the PyTorch automatic-differentiation implementation is separately tested against the symbolic residual. The corresponding equations printed in the source paper (its Eqs. (37)–(38)) fail the same numeric identity check: they attach the β_3 and β_4 terms to the wrong field components and leave the complex coefficient $(\gamma + i\alpha_2/2)$ outside $\text{Re}[\cdot]/\text{Im}[\cdot]$.

S2 Audit methodology

Ansatz and certificate. The paper’s ansatz is $A = aJ_{n,m}(\xi)$, $\xi = \lambda_1 z + \lambda_2 t$, $J_{n,m} = \sinh^m \xi / \cosh^n \xi$, with a real and no carrier phase. On $\xi > 0$ we set $T = \tanh \xi \in (0, 1)$ and $W = \text{sech } \xi$. Then $J_{n,m} = T^m W^{n-m}$, the logarithmic derivatives $J^{(k)}/J$ are rational functions of T , and $J^2 = T^{2m}(1 - T^2)^{n-m}$. Substituting the ansatz into Eq. (1), dividing by A , and clearing only powers of T and $1 - T^2$ — both nonzero on the open domain — converts exactness into the vanishing of all coefficients of a polynomial identity in T (for half-odd-integer $n - m$, additionally split over the basis $\{1, W\}$, which is linearly independent over the rational functions of T). This certificate is exhaustive; in particular it does not assume distinct J -indices to be linearly independent, an assumption that underlies the paper’s classification and is false ($J_{2,0} = J_{0,0} - J_{2,2}$, i.e. $\text{sech}^2 = 1 - \tanh^2$).

Complete enumeration of closures. A bounded recursive factor-splitting solver enumerates every solution family of each closure system over the unknowns $(\lambda_1, \lambda_2, a^2, \beta_2, \beta_3, \beta_4, c, \ell, \gamma, \alpha_2, \Gamma)$, refusing $a = 0$ and flagging $\lambda_2 = 0$ as degenerate. Every terminal family is then verified through two independent gates:

1. **Symbolic gate:** the family’s triangular substitutions are inserted into the certificate with all remaining free parameters kept symbolic; every coefficient must simplify to exactly zero (this proves the whole family at once).
2. **Numeric gate:** exact rational representative parameter sets are substituted into the *original complex* PDE (1) and the residual is evaluated at 50-digit precision on a dense ξ -grid plus random (z, t) points, with acceptance thresholds $\max|F| < 10^{-10}$ and $\|F\|_2/\|A\|_2 < 10^{-10}$ (both configurable; all accepted families in fact reach 10^{-47} or better).

Degenerate candidates are rejected on explicit grounds: $a = 0$; $\lambda_2 = 0$; identically vanishing governing equation (at most one active PDE term on the family — “vacuous”); denominator singularities inside the declared domain; residual failure away from any fitting points. Fractional powers of $\sinh$ are never interpreted silently: the implementation distinguishes the principal complex power, a signed-real convention, and the $\xi > 0$ restriction, and profiles that are not C^4 at $\xi = 0$ (the PDE contains A_{ttt}) are either domain-restricted or rejected.

Classification findings. Solving the paper’s own index-matching requirement $J_{n+r,m+r} = J_{3n+s,3m+s}$ for both indices simultaneously forces $n = m$ (its Eq. (28)); the 42 off-diagonal pairs of its 7×7 catalogue are therefore unsupported by its own argument. The audit shows the classification is also not *necessary*: the only bright branch that genuinely closes, $(1, 0) = \text{sech}$, is off-diagonal.

S3 The verified catalogue

Of the 49 proposed pairs, **9 admit at least one verified nontrivial solution family; 40 admit only vacuous or $\lambda_2 = 0$ closures** (per-pair reasons in `data/verified_branches/rejected_pairs.csv`). Every surviving family is a *reduced* model: the closure forces several physical coefficients to zero, and no family supports the full nine-coefficient equation (no Tier A solutions under the paper’s constant-phase convention). Table S1 lists the accepted families; Fig. S1 shows the three benchmark-grade profiles.

Table S1: Verified solution families under the paper’s constant-phase, real-amplitude ansatz. “#coeffs forced 0” counts physical coefficients that the closure sets to zero. Residuals are 50-digit evaluations of Eq. (1) on dense and random points.

(n, m)	profile	tier	group	#coeffs forced 0	$\max F $	rel. RMS
(0, 0)	constant	B	main	1	0	0
(1, 0)	bright sech	B	main	5	1.3×10^{-51}	5.9×10^{-52}
(1, 1)	dark tanh kink	B	main	5	2.0×10^{-51}	6.0×10^{-52}
(−1, −1)	coth (singular)	B	stress	5	2.1×10^{-50}	2.4×10^{-51}
(−1, 0)	cosh (unbounded)	B	stress	2	2.1×10^{-50}	4.8×10^{-52}
(−1, 1)	$\frac{1}{2} \sinh(2\xi)$ (unbounded)	B	stress	2	3.8×10^{-48}	2.5×10^{-51}
(0, −1)	csch (singular)	B	stress	5	2.1×10^{-50}	3.7×10^{-51}
(0, 1)	$\sinh$ (unbounded)	B	stress	2	2.0×10^{-50}	4.7×10^{-52}
(1, −1)	$2 \operatorname{csch}(2\xi)$ (singular)	B	stress	5	4.3×10^{-50}	8.4×10^{-51}

The closures of the three main-group branches are, explicitly:

$$(0, 0) : \quad \gamma = 0, \quad \ell = \frac{\alpha_2 a^2}{2(1 + \Gamma a^2)} \quad (\text{saturation } \Gamma > 0 \text{ admitted; } \beta_{2,3,4} \text{ free}) \quad (5)$$

$$(1, 0) : \quad \beta_2 = \beta_3 = \beta_4 = \gamma = \Gamma = 0, \quad \lambda_1 = 0, \quad \alpha_2 a^2 = 2c\lambda_2^2, \quad \ell = \frac{1}{2}c\lambda_2^2, \quad (6)$$

$$(1, 1) : \quad \beta_2 = \beta_3 = \beta_4 = \gamma = \Gamma = 0, \quad \lambda_1 = 0, \quad \alpha_2 a^2 = -2c\lambda_2^2, \quad \ell = -c\lambda_2^2. \quad (7)$$

Physically, the sech branch is a stationary dissipative bright pulse (net linear gain balanced by two-photon absorption and spectral filtering) and the kink is its loss-side mirror (net linear loss balanced by nonlinear gain). Notably, *no nonconstant branch tolerates saturation* $\Gamma > 0$, and the Kerr coefficient γ is forced to zero on every family — consequences of the constant-phase convention examined in Section S4.

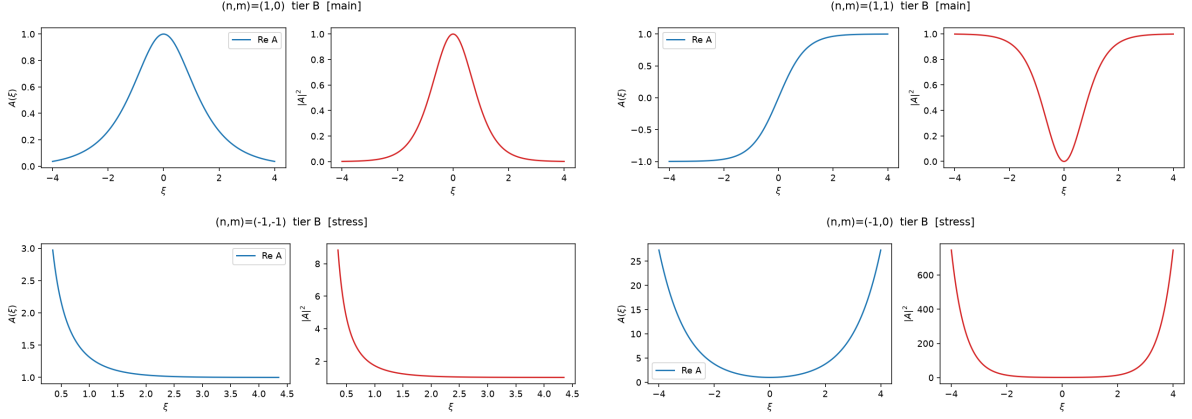


Figure S1: Verified profiles at their first representative parameter sets. Top: the two nonconstant main-group branches — bright sech (1, 0) and dark kink (1, 1). Bottom: two stress-group examples — the singular coth branch (−1, −1) on its restricted domain $\xi \in [0.35, 4.35]$, and the unbounded cosh branch (−1, 0), the latter a solution of the *linear* reduced equation with third- and fourth-order dispersion active and a genuinely travelling frame ($\lambda_1 = -\beta_3 \lambda_2^3/6$).

S4 Optional extended-phase audit

As a clearly separated experiment (never a replacement of the paper’s formulation), the full 49-pair audit was repeated for the carrier ansatz $A = a J_{n,m}(\xi) e^{i(\kappa z + \omega t)}$. The same two verification gates apply; all accepted extended families reach residuals $\approx 10^{-51}$. Results: **17 verified families, all on the same nine pairs — none of the 40 rejected pairs is rescued.** The carrier restores exactly the physics the constant-phase convention destroys:

- on (1, 0), the classic *travelling bright NLS soliton*: $\gamma = \beta_2 \lambda_2^2/a^2$ (focusing), with κ the dispersion relation and $\lambda_1 = \beta_2 \lambda_2 \omega$ the group-velocity shift, in the conservative reduction $c = \ell = \alpha_2 = 0$;
- on (1, 1), the *travelling dark soliton*: $\gamma = -\beta_2 \lambda_2^2/a^2$ (defocusing);
- on (0, 0), a **Tier-A continuous-wave solution** with saturation, fourth-order dispersion and gain/loss simultaneously active — the only full-model solution found anywhere in this study.

Conclusion: the catalogue’s collapse is caused by the paper’s “*a* real without loss of generality” assumption, not by the equation itself.

S5 Independent numerical validation

Each main-group solution was propagated in z by a conventional method-of-lines solver, entirely independent of the symbolic machinery: sixth-order central finite differences in t with three ghost nodes per side filled from the exact analytic boundary data (kinks are never given periodic boundary conditions), integrated with DOP853; for periodic-compatible profiles (localized or constant) a Fourier pseudospectral discretization with an ETDRK4 exponential integrator on a $3\times$ padded window provides a second, structurally different cross-check. The convergence study covers three spatial resolutions and two integrator tolerances (Table S2).

Agreement at the 10^{-11} – 10^{-13} level between analytically verified solutions and an unrelated numerical method confirms the catalogue independently before any network training.

Table S2: Convergence study of the independent solver. The finite-difference scheme exhibits its design order (≈ 6 ; the constant branch is at the rounding floor, where measured “orders” are meaningless). The ETDRK4 cross-check on the sech branch is limited to $\sim 10^{-6}$ by periodic tail truncation of the non-periodic profile, as expected.

branch	scheme	n_t	tol / steps	final L2RE	max err	runtime (s)
constant (0, 0)	fd_mol	65	1.0×10^{-8}	9.9×10^{-12}	1.4×10^{-11}	0.04
	fd_mol	65	1.0×10^{-10}	1.6×10^{-13}	2.4×10^{-13}	0.04
	fd_mol	129	1.0×10^{-8}	8.4×10^{-12}	1.2×10^{-11}	0.73
	fd_mol	129	1.0×10^{-10}	2.5×10^{-13}	4.5×10^{-13}	0.74
	fd_mol	257	1.0×10^{-8}	5.3×10^{-11}	7.5×10^{-11}	13.47
	fd_mol	257	1.0×10^{-10}	1.2×10^{-12}	1.6×10^{-12}	13.42
	ETDRK4	256	400 steps	8.6×10^{-18}	8.6×10^{-18}	0.06
observed spatial orders: -0.6, -2.3						
sech (1, 0)	fd_mol	65	1.0×10^{-8}	2.3×10^{-7}	4.4×10^{-7}	0.01
	fd_mol	65	1.0×10^{-10}	2.3×10^{-7}	4.4×10^{-7}	0.02
	fd_mol	129	1.0×10^{-8}	7.6×10^{-9}	1.2×10^{-8}	0.05
	fd_mol	129	1.0×10^{-10}	3.7×10^{-9}	7.2×10^{-9}	0.05
	fd_mol	257	1.0×10^{-8}	4.6×10^{-10}	5.0×10^{-10}	0.20
	fd_mol	257	1.0×10^{-10}	5.8×10^{-11}	1.2×10^{-10}	0.21
	ETDRK4	256	400 steps	2.3×10^{-6}	5.3×10^{-6}	0.06
observed spatial orders: 6.0, 6.0						
kink (1, 1)	fd_mol	65	1.0×10^{-8}	1.1×10^{-7}	3.1×10^{-7}	0.01
	fd_mol	65	1.0×10^{-10}	1.1×10^{-7}	3.1×10^{-7}	0.02
	fd_mol	129	1.0×10^{-8}	1.8×10^{-9}	5.2×10^{-9}	0.05
	fd_mol	129	1.0×10^{-10}	1.8×10^{-9}	5.2×10^{-9}	0.05
	fd_mol	257	1.0×10^{-8}	3.7×10^{-9}	5.0×10^{-9}	0.21
	fd_mol	257	1.0×10^{-10}	7.5×10^{-11}	1.0×10^{-10}	0.21
	observed spatial orders: 6.0, 4.6					

S6 Forward PINN benchmark

Setup. Networks map $(z, t) \mapsto (\hat{u}, \hat{v})$, $\hat{A} = \hat{u} + i\hat{v}$; all computations in `float64`; tanh activations (smooth through fourth derivatives); inputs affinely scaled to $[-1, 1]$. Training data comprise exact initial values at $z = 0$ and exact Dirichlet values at the two t -boundaries only — *no interior solution values* — plus the PDE residual (Eqs. (3)–(4)) at Latin-hypercube collocation points. Errors are evaluated on an independent uniform grid, and residuals on points never used in training. Three methods: fixed loss weights ($w_{IC} = w_{BC} = 100$, $w_{PDE} = 1$); gradient-norm adaptive weighting (Wang et al. 2021); residual-adaptive collocation (RAD, Wu et al. 2023). Pilot budget on the available hardware (4-core CPU, no GPU): 4×40 network, 1024 collocation points, 600 Adam + 100 L-BFGS steps (the recommended GPU configuration — 6×64 , 10^4 points — ships unchanged in `configs/full.yaml`). Matrix: 3 branches $\times$ 2 closure-respecting parameter sets $\times$ 3 methods $\times$ 5 seeds = 90 runs, **zero failures**. Success criterion (configurable): complex L2RE $< 10^{-2}$ together with a finite independently sampled residual, where $\text{L2RE} = (\sum_j |\hat{A}_j - A_j|^2 / \sum_j |A_j|^2)^{1/2}$.

Table S3: Complete pilot matrix, pooled over the two parameter sets ($n = 10$ runs per cell). No selection of favourable seeds: minima and maxima are shown.

branch	method	n	median	mean $\pm$ std	range	success	resid. RMS
constant (0, 0)	fixed weights	10	5.3×10^{-4}	$9.7 \times 10^{-4} \pm 7.8 \times 10^{-4}$	$3.4 \times 10^{-4} - 2.3 \times 10^{-3}$	100%	2.2×10^{-3}
	adaptive weights	10	3.8×10^{-3}	$4.1 \times 10^{-3} \pm 3.1 \times 10^{-3}$	$2.6 \times 10^{-4} - 8.6 \times 10^{-3}$	100%	6.6×10^{-3}
	adaptive sampling	10	1.3×10^{-3}	$1.6 \times 10^{-3} \pm 1.1 \times 10^{-3}$	$2.6 \times 10^{-4} - 3.4 \times 10^{-3}$	100%	3.5×10^{-3}
bright sech (1, 0)	fixed weights	10	1.9×10^{-2}	$1.9 \times 10^{-2} \pm 3.3 \times 10^{-3}$	$1.4 \times 10^{-2} - 2.4 \times 10^{-2}$	0%	2.6×10^{-2}
	adaptive weights	10	1.3×10^{-2}	$1.5 \times 10^{-2} \pm 8.4 \times 10^{-3}$	$4.5 \times 10^{-3} - 2.9 \times 10^{-2}$	50%	1.9×10^{-2}
	adaptive sampling	10	1.6×10^{-2}	$1.7 \times 10^{-2} \pm 3.6 \times 10^{-3}$	$1.0 \times 10^{-2} - 2.4 \times 10^{-2}$	0%	2.4×10^{-2}
dark kink (1, 1)	fixed weights	10	5.3×10^{-3}	$4.8 \times 10^{-3} \pm 1.1 \times 10^{-3}$	$2.9 \times 10^{-3} - 6.1 \times 10^{-3}$	100%	1.8×10^{-2}
	adaptive weights	10	2.4×10^{-3}	$3.1 \times 10^{-3} \pm 1.7 \times 10^{-3}$	$1.3 \times 10^{-3} - 5.9 \times 10^{-3}$	100%	9.3×10^{-3}
	adaptive sampling	10	6.0×10^{-3}	$5.5 \times 10^{-3} \pm 1.5 \times 10^{-3}$	$3.2 \times 10^{-3} - 7.4 \times 10^{-3}$	100%	1.9×10^{-2}

Findings. (i) Difficulty is ordered exactly as predicted by the catalogue’s shape features: constant $<$ kink $<$ sech, tracking the maxima of $|A_{tt}|$ and, dominantly, $|A_{tttt}|$ — the operative feature given the equation’s fourth-derivative term (Fig. S6). (ii) Adaptive loss weighting is the only method to lift the sech branch to 50% success at pilot budget, while *degrading* the easiest branch (it deprioritizes already-converged data terms); adaptive collocation helps the constant branch and is neutral elsewhere.

S6.1 High-fidelity rerun of the failing branch

To decide whether the sech failure is fundamental or budget-limited, the branch was retrained at $2.5\times$ the optimization steps and $2\times$ the collocation points (4×48 network; 18 runs). Median L2RE falls from 1.6×10^{-2} to 3.5×10^{-3} with **83% overall success** (Table S4): the pilot-budget failure is largely a *budget artifact*. The rerun also reverses the method ranking — fixed weights and adaptive sampling reach 100% success while adaptive weighting drops to 50%: **method rankings are budget-dependent**, a caution against comparing PINN variants at a single training budget.

S6.2 Stress group

The six stress branches — singular profiles on domains bounded away from $\xi = 0$, and unbounded cosh/sinh-type solutions of the linear reduced equation with β_3, β_4 active and travelling frames

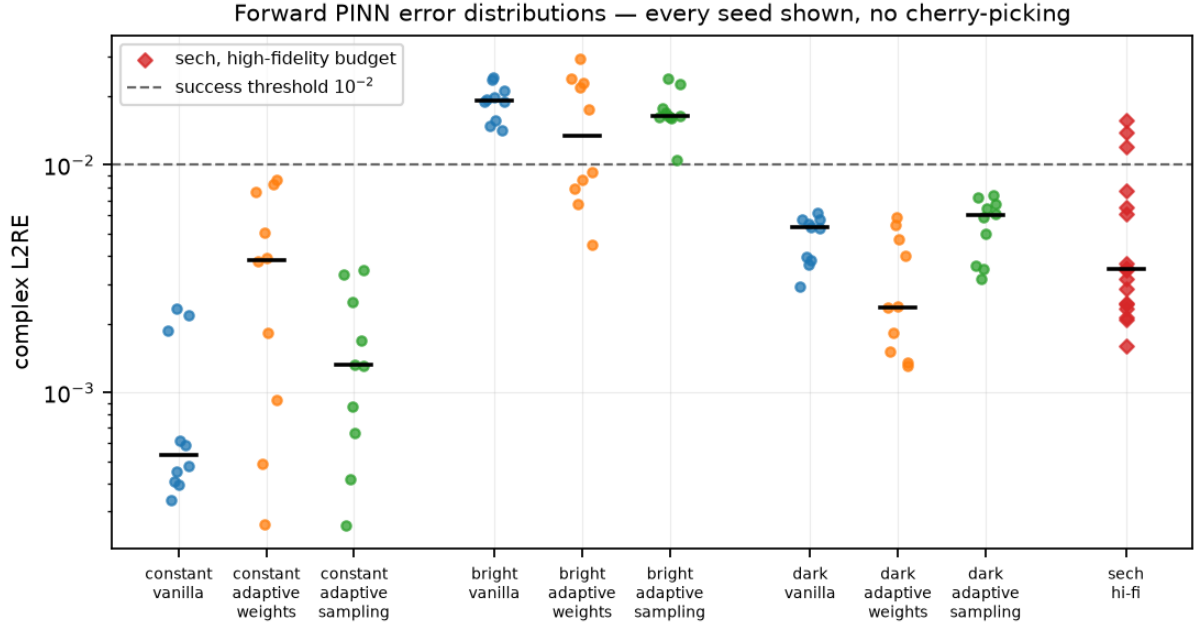


Figure S2: Every run of the pilot matrix (dots), medians (bars), and the high-fidelity sech runs (red diamonds). The constant and kink branches sit one to two decades below the 10^{-2} threshold for every method; the bright sech branch straddles it at pilot budget.

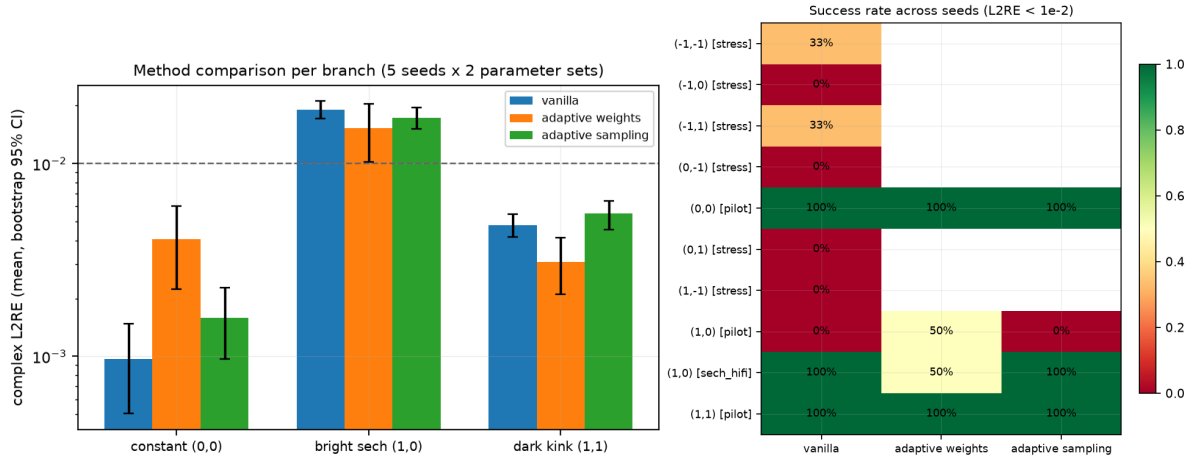


Figure S3: Left: per-branch method comparison with bootstrap 95% confidence intervals. Right: success rates across seeds for all forward experiments, including the stress group and the high-fidelity sech rerun.

Table S4: High-fidelity sech runs (2500 Adam + 300 L-BFGS steps, 2048 points).

method	n	median L2RE		range	success
fixed weights	6	3.5×10^{-3}	$2.1 \times 10^{-3} - 6.5 \times 10^{-3}$		100%
adaptive weights	6	7.8×10^{-3}	$1.6 \times 10^{-3} - 1.6 \times 10^{-2}$		50%
adaptive sampling	6	2.7×10^{-3}	$2.1 \times 10^{-3} - 7.7 \times 10^{-3}$		100%
all methods	18	3.5×10^{-3}	$1.6 \times 10^{-3} - 1.6 \times 10^{-2}$		83%

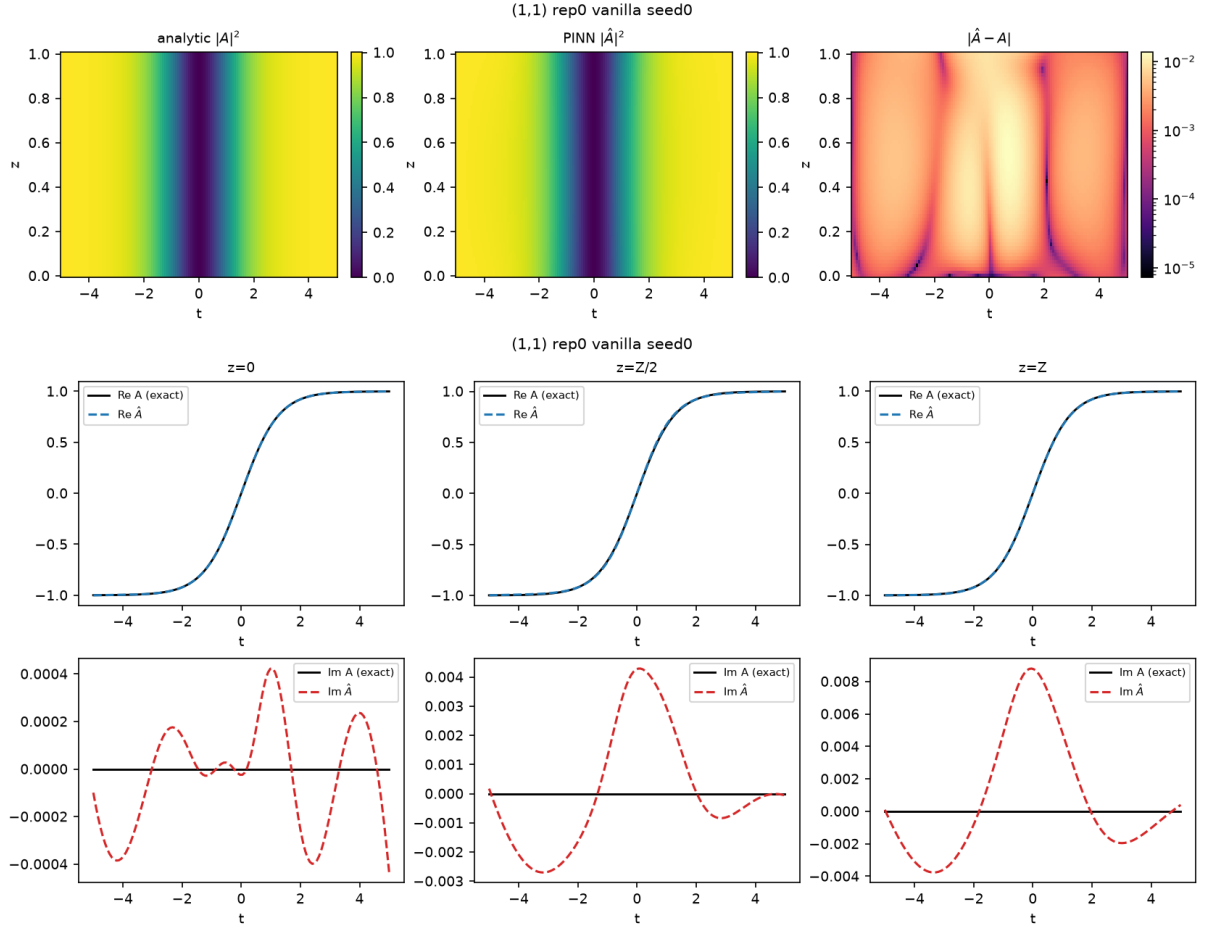


Figure S4: Representative reconstruction (dark kink, fixed weights, median seed). Top: analytic intensity $|A|^2$, PINN intensity, and pointwise complex error. Bottom: real and imaginary components at $z = 0, Z/2, Z$; the network must also learn that $\text{Im } A \equiv 0$.

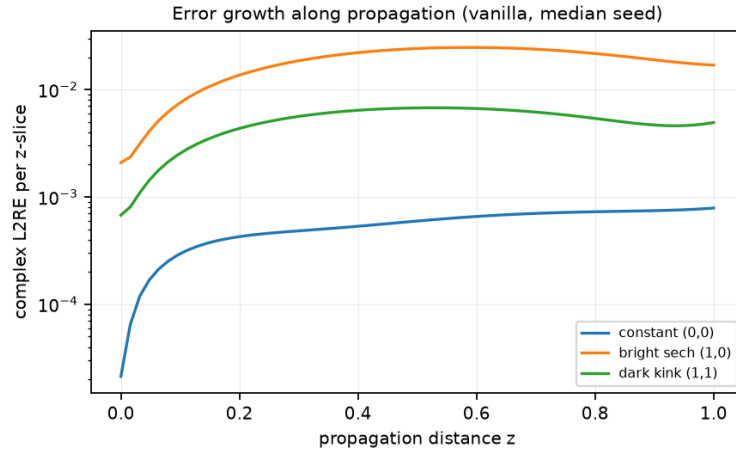


Figure S5: Per-slice error along the propagation direction (fixed weights, median seeds). Errors are anchored by the exact data at $z = 0$ and grow into the interior, where the residual term alone carries the solution.

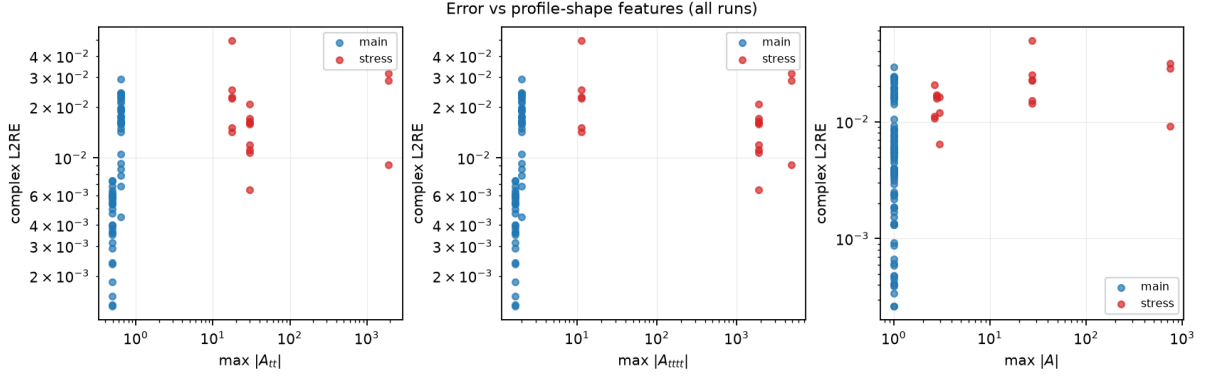


Figure S6: Error against profile-shape features for all forward runs (main and stress groups). The correlation with the fourth-derivative magnitude is the strongest of the three.

— were trained for the first time at pilot budget (18 runs, fixed weights, Table S5). All six sit at or above the success threshold (medians $1.1\text{--}2.9 \times 10^{-2}$, success 0–33%): near-singular gradients, field amplitudes up to ~ 27 , and active high-order dispersion each cost one to two decades relative to the main group. These branches delimit the failure regime of the benchmark and are reported without selection.

Table S5: Stress-group results (fixed weights, 3 seeds per branch).

(n, m)	profile	n	median L2RE	range	success
$(-1, -1)$	coth kink (singular)	3	1.2×10^{-2}	$6.4 \times 10^{-3} - 1.6 \times 10^{-2}$	33%
$(-1, 0)$	cosh (unbounded, β_3, β_4 active)	3	2.5×10^{-2}	$2.3 \times 10^{-2} - 4.9 \times 10^{-2}$	0%
$(-1, 1)$	$\frac{1}{2} \sinh(2\xi)$ (unbounded, travelling)	3	2.9×10^{-2}	$9.1 \times 10^{-3} - 3.2 \times 10^{-2}$	33%
$(0, -1)$	csch pulse (singular)	3	1.6×10^{-2}	$1.6 \times 10^{-2} - 1.7 \times 10^{-2}$	0%
$(0, 1)$	sinh (unbounded, travelling)	3	1.5×10^{-2}	$1.4 \times 10^{-2} - 2.3 \times 10^{-2}$	0%
$(1, -1)$	$2 \operatorname{csch}(2\xi)$ (singular)	3	1.1×10^{-2}	$1.1 \times 10^{-2} - 2.1 \times 10^{-2}$	0%

S7 Inverse parameter discovery

Identifiability first. Because g_0, T_2, α enter Eq. (1) only through c and ℓ , they cannot be separately recovered from any single field. The sensitivity Jacobian $\partial F / \partial \theta$ evaluated on the exact solutions confirms this numerically: rank 2 of 3 in (g_0, T_2, α) with an explicit null direction, and rank 7 of 8 with condition number $\sim 10^{11}$ in the reduced coordinates (Table S6). All recovery is therefore performed in $\theta = (\beta_2, \beta_3, \beta_4, c, \ell, \gamma, \alpha_2, \Gamma)$, with positive parameters (c, Γ) in softplus parameterization and all parameters scaled to $O(1)$.

Table S6: Sensitivity analysis on the exact solutions (400 random points). The near-zero third singular value in (g_0, T_2, α) and its null direction prove joint non-identifiability; recovery targets c and ℓ instead.

branch	reduced rank	cond. number	(g_0, T_2, α) rank	null direction
kink (1, 1)	7/8	7.6×10^{10}	2/3	$(-0.686, 0.243, -0.686)$
sech (1, 0)	7/8	1.1×10^{11}	2/3	$(-0.686, 0.243, -0.686)$

Staged recovery. Network and parameters are trained jointly from sparse noisy interior observations plus the residual (1200 Adam steps; initial parameter guesses offset by $\pm 50\%$; five seeds configured, three executed per cell on this hardware; both an easy branch — the kink —

and the hardest — the sech). Noise is Gaussian with standard deviation quoted relative to the field’s standard deviation. Results: Table S7 and Fig. S7.

Table S7: Median recovery error at 5% observation density, by stage and noise level (both branches pooled; median over seeds). For parameters whose true value is 0 the entry is the *absolute* error — these rows measure the ability to *discover* that a term is absent.

stage	parameter	0% noise	1% noise	5% noise	runs
gainloss	$c = g_0 T_2^2$	1.8×10^{-2}	3.0×10^{-2}	6.4×10^{-2}	6
	$\ell = (g_0 - \alpha)/2$	3.3×10^{-3}	7.3×10^{-3}	2.2×10^{-2}	6
nonlinear	γ (true 0)	1.5×10^{-3}	7.3×10^{-4}	2.5×10^{-3}	6
	Γ (true 0)	2.3×10^{-2}	2.4×10^{-2}	3.0×10^{-2}	6
	α_2	1.5×10^{-2}	1.2×10^{-2}	2.4×10^{-2}	6
dispersion	β_2 (true 0)	9.1×10^{-3}	6.9×10^{-3}	1.8×10^{-2}	6
	β_3 (true 0)	2.6×10^{-2}	2.1×10^{-2}	7.2×10^{-2}	6
	β_4 (true 0)	4.3×10^{-2}	6.6×10^{-2}	7.7×10^{-2}	6

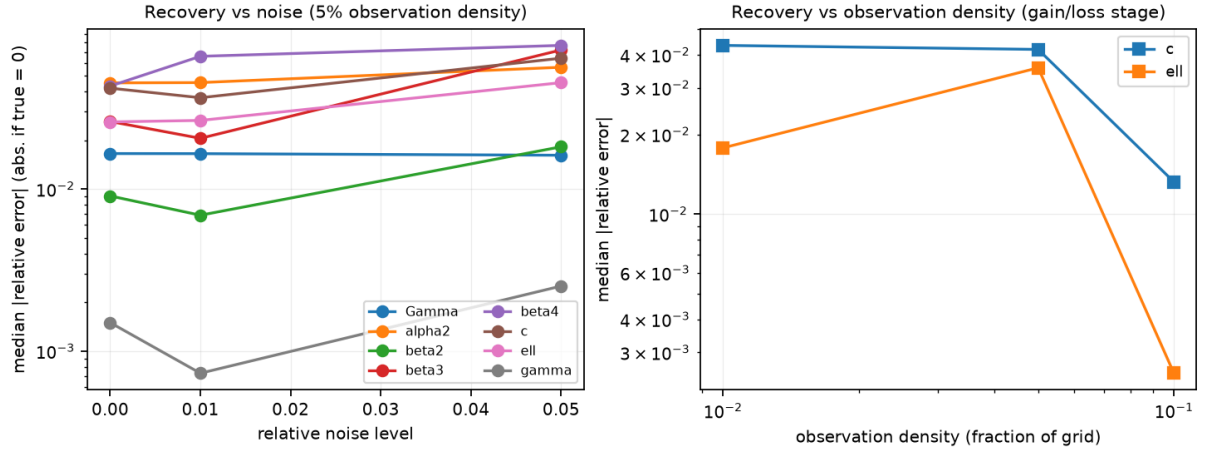


Figure S7: Left: median recovery error against noise at 5% density. Right: recovery of the gain/loss combinations against observation density (1%, 5%, 10%) — a clean monotone improvement.

Table S8: Observation-density sweep for the gain/loss stage (noise levels pooled; median over seeds and branches).

parameter	1% density	5% density	10% density
$c = g_0 T_2^2$	4.3×10^{-2}	3.0×10^{-2}	1.3×10^{-2}
$\ell = (g_0 - \alpha)/2$	1.8×10^{-2}	7.6×10^{-3}	2.5×10^{-3}

Findings. (i) The identifiable gain/loss combinations are the best-recovered quantities (ℓ to 0.3–2%, c to 1–7% across 0–5% noise), degrading gracefully with noise and improving monotonically with density (Table S8). (ii) The nonlinear stage recovers α_2 to 1–3% and correctly discovers $\gamma \approx 0$ and $\Gamma \approx 0$. (iii) The dispersion stage is the weakest: the β ’s (true value 0 on these branches) are pinned only to median absolute error 3.3×10^{-2} — high-order derivative terms have weak sensitivity on smooth solutions. (iv) Joint recovery is much harder than staged recovery (Table S9). (v) A structural consequence of the audit: on the surviving branches most coefficients are *forced to zero*, so the paper’s proposed nine-parameter recovery benchmark largely reduces to discovering zeros.

Table S9: Staged versus joint recovery. Joint four-parameter recovery degrades by roughly an order of magnitude for the genuinely nonzero parameters, consistent with the $\sim 10^{11}$ conditioning of the sensitivity matrix.

parameter	staged (median)	joint 4-param (median)	joint runs
$c = g_0 T_2^2$	3.0×10^{-2}	1.2×10^{-1}	18
$\ell = (g_0 - \alpha)/2$	7.6×10^{-3}	1.6×10^{-1}	18
α_2	1.7×10^{-2}	1.5×10^{-1}	18
Γ (true 0)	2.3×10^{-2}	1.5×10^{-2}	18

S8 Physical ablations

Matched pairs change exactly one closure-consistent quantity on the constant branch — the only branch whose closure admits them (Table S10). Saturation ($\Gamma : 0 \rightarrow 1$) and fourth-order dispersion in the operator ($\beta_4 : 0 \rightarrow 0.4$) each make the forward problem mildly harder ($\times 1.2$ – 1.4 in mean L2RE). The remaining requested ablations are *impossible by closure*, which is itself a reportable finding: no verified branch admits a conservative version ($\ell = \alpha_2 = 0$ collapses every family), and $\Gamma > 0$ or $\beta_4 \neq 0$ are closure-forbidden off the constant branch — the verified solution set is intrinsically dissipative. Fixed-versus-adaptive weighting and uniform-versus-adaptive sampling are covered by the factorial matrix of Section S6.

Table S10: Matched ablations (3 seeds per arm, fixed weights, 800 Adam + 150 L-BFGS steps).

arm (constant branch, one closure-consistent change)	mean L2RE $\pm$ std	n
saturation off ($\Gamma = 0$)	$6.1 \times 10^{-4} \pm 2.1 \times 10^{-4}$	3
saturation on ($\Gamma = 1$)	$8.8 \times 10^{-4} \pm 3.0 \times 10^{-4}$	3
$\beta_4 = 0$	$9.2 \times 10^{-4} \pm 3.5 \times 10^{-4}$	3
$\beta_4 = 0.4$	$1.1 \times 10^{-3} \pm 6.7 \times 10^{-4}$	3

S9 Parameter-conditioned cross-branch PINN

A network taking only (z, t) cannot represent multiple solutions at the same coordinates, so generalization experiments use $\hat{A}_\theta(z, t; \eta)$ with $\eta = (n, m, a, \lambda_1, \lambda_2, \beta_2, \beta_3, \beta_4, c, \ell, \gamma, \alpha_2, \Gamma)$ scaled over the task pool. Training tasks are the catalogue representatives of the training branches; evaluation tasks at new parameters are generated *only* by re-solving the exact closure constraints at new free-parameter values, never by free perturbation. Train/held-out branch lists are explicit in `configs/generalization_pilot.yaml`; in leave-one-branch-out (LOBO) runs the held-out branch contributes nothing to training, and fine-tuning uses only the stated fraction of held-out interior data (two seeds).

Table S11: Conditioned-PINN transfer, mean L2RE over seeds and the two held-out parameter sets. “Scratch, same budget” trains a fresh single-branch network for exactly the fine-tuning step budget.

held-out branch	zero-shot	fine-tune 1%	fine-tune 5%	scratch, same budget
constant (0, 0)	7.1×10^{-1}	8.3×10^{-3}	6.2×10^{-3}	1.1×10^{-2}
sech (1, 0)	1.3×10^0	6.7×10^{-2}	4.8×10^{-2}	9.9×10^{-2}
kink (1, 1)	1.1×10^0	6.0×10^{-2}	2.5×10^{-2}	5.8×10^{-2}
interpolation (all branches): mean L2RE 1.3×10^{-1}				
extrapolation (all branches): mean L2RE 6.7×10^{-1}				

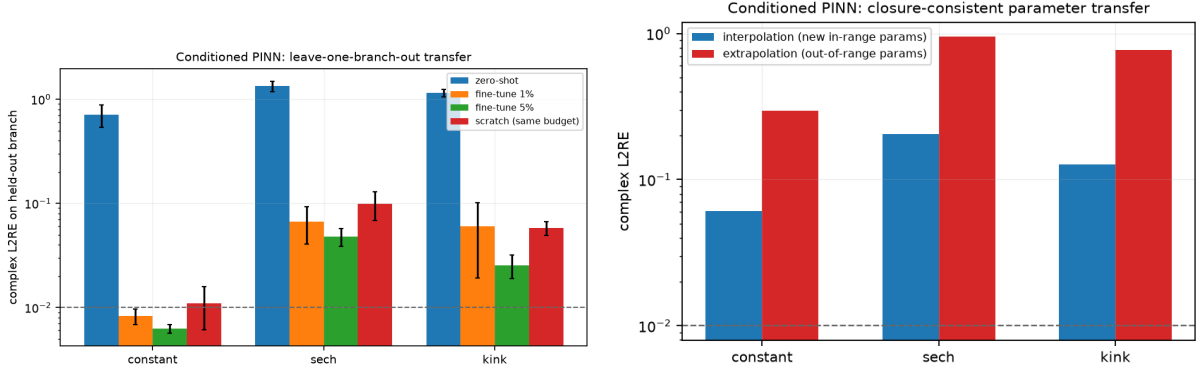


Figure S8: Left: LOBO transfer — zero-shot fails ($L2RE \approx 1$); 1% of held-out data recovers scratch-level or better accuracy at matched compute. Right: transfer to new closure-consistent parameters — interpolation partially succeeds, extrapolation fails.

Findings. Zero-shot transfer to an unseen branch fails completely ($L2RE \approx 0.7$ – 1.3): the conditioning vector does not let the network infer a qualitatively different profile shape it has never seen — an expected but now quantified negative result. Interpolation to unseen in-range parameters of *seen* branches partially succeeds (mean 1.3×10^{-1}); extrapolation fails (mean 6.7×10^{-1}). Fine-tuning with 1% of held-out data restores accuracy comparable to — on two of three branches better than — a from-scratch network at the same step budget, so the shared representation carries some reusable structure.

S10 Reproducibility, compute, and honest scope

Environment. Linux, 4-core CPU, no GPU; Python 3.12; PyTorch (CPU, `float64`), SymPy/mpmath for the audit (50-digit arithmetic); all package versions logged per experiment in `results/*/environment.json`. Seeds are fixed and recorded; per-run raw results are archived before aggregation; checkpointing allows every experiment to resume, and completed jobs are skipped unless `--force` is passed. The complete unit-test suite (42 tests: symbolic derivatives, residual-split identity, autodiff-versus-symbolic derivatives, fourth-derivative accuracy, fractional-power conventions, singularity detection, closure verification, degenerate rejection, metrics, leakage, deterministic seeding, checkpoint/resume, end-to-end smoke) passes before and after all experiments.

Compute scale. Roughly 230 training runs are archived: 90 pilot, 18 high-fidelity sech, 18 stress, 108 inverse, plus conditioned-PINN trainings and matched ablations. Network and collocation sizes are the documented CPU scaling of the recommended settings; the single configuration not executed end-to-end on this machine is the GPU-scale `configs/full.yaml` (6×64 , 10^4 collocation points, 5000+500 steps), which is expected to move absolute errors down without changing the qualitative orderings established here — the high-fidelity sech rerun (Section S6.1) is direct evidence for that trend. Inverse cells ran 3 of the 5 configured seeds.

Commands.

```
python scripts/run_audit.py --config configs/audit.yaml    # +---extended-phase
python -m pytest tests/ -q
python scripts/run_numerical.py
python scripts/run_forward.py --config configs/pilot.yaml
python scripts/run_forward.py --config configs/sech_highfidelity.yaml
python scripts/run_forward.py --config configs/stress.yaml
python scripts/run_inverse.py --config configs/inverse_pilot.yaml
```

```
python scripts/run_ablations.py
python scripts/run_generalization.py --config configs/generalization_pilot.yaml
python scripts/run_full_benchmark.py --mode smoke|pilot|full    # whole pipeline
```

Provenance of claims. Mathematically proven: every family in `branch_catalog.json` (symbolic zero + 50-digit residuals). Unsupported: the 40 rejected pairs, with per-pair reasons. PINN successes and failures: per-run `result.json` files under `results/`; raw errors are always retained and no seed is excluded anywhere in this document.